Electrolytic cell and electrolysis

Electrolysis is a process in which the electrical energy is used to cause a non-spontaneous chemical reaction to occur; the energy is often used to decompose a compound into its elements. The device which is used to carry out the electrolysis is called the electrolytic cell. The electrochemical process occurring in the electrolytic cell and galvanic cell are the reverse of each other. Let us understand the function of a electrolytic cell by considering the electrolysis of molten sodium chloride.

The electrolytic cell consists of two electrodes one is cylindrical steel cathode and another one is graphite anode. They are dipped in molten sodium chloride. They are connected to the external DC power supply via a key as shown in the figure (9.8). The electrode which is attached to the negative end of the power supply is called the cathode, and the one which is attached to the positive end is called the anode. Once the key is closed, the external DC power supply drives the electrons to the cathode and at the same time pull the electrons from the anode.

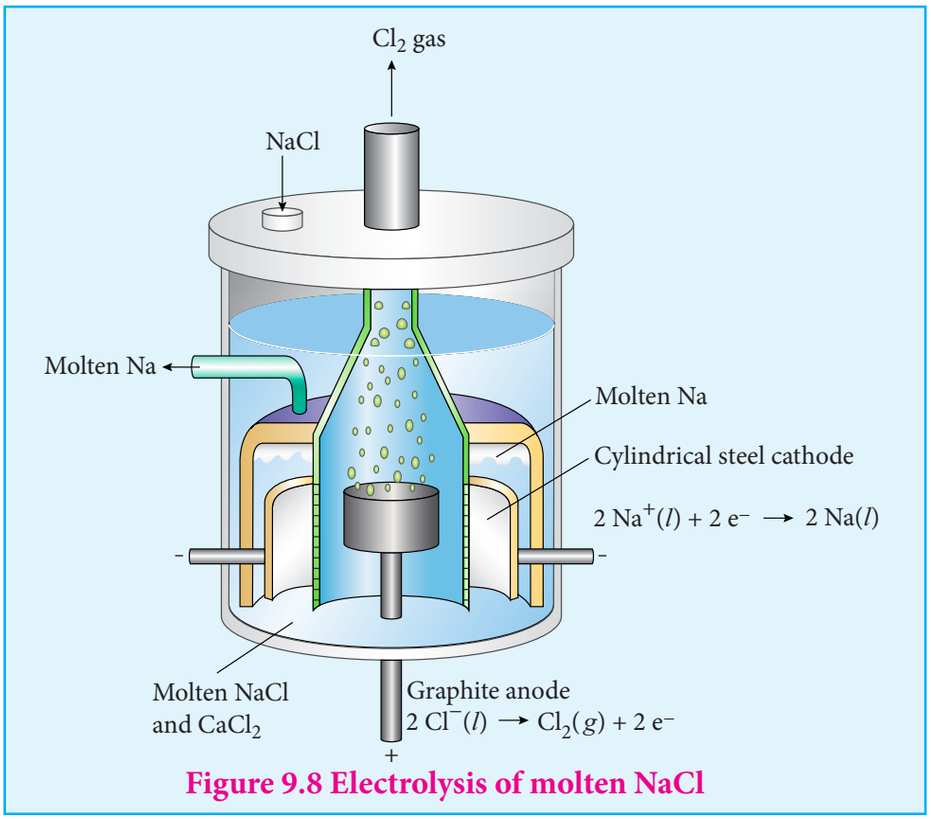


Figure 9.8 Electrolysis of molten NaCl

Cell reactions

Na^+ ions are attracted towards cathode, where they combine with the electrons and reduced to liquid sodium.

Cathode (reduction)

$Na^+(l) + e^- \rightarrow Na(l)$ $\qquad E^\circ = -2.71V$

Similarly, Cl^- ions are attracted towards anode where they lose their electrons and oxidised to chlorine gas.

Anode (oxidation)

$2Cl^-(l) \rightarrow Cl_2(g) + 2e^-$ $\qquad E^\circ = -1.36V$

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